JJJ-S-581c
December 14, 1965

SUPERSEDING
Int. Fed. Spec. JJJ-S-00581b (Army-GL)
December 21, 1964 and
Fed. Spec. JJJ-S-581a
July 25, 1955

FEDERAL SPECIFICATION

SOUPS; CANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers canned soups for use by agencies of the Federal Government.

1.2 Classification. Canned soups shall be of the following styles, types, classes, and forms, as specified (see 6.1):

Style A—Standard market varieties (type and class, as specified) (see 6.1).
Form a—Ready to serve.
Form b—Condensed.
Style B—Special varieties.
Type I—Vegetable with beef.
Type II—Chicken.
Class 1—Noodle.
Class 2—Rice.
Type III—Cream.
Class 1—Tomato.
Class 2—Mushroom.
Class 3—Pea.
Class 4—Chicken.
Form a—Ready to serve.
Form b—Condensed.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Specifications:

C-C-671—Cream and Half-and-Half, Fresh.
C-M-350—Milk, Nonfat, Dry.
C-M-371—Milk, Evaporated.
C-M-381—Milk, Whole, Fresh.
N-B-121—Barley, Pearl.
N-C-541—Cornstarch; Edible.
N-F-481—Flour, Wheat.
N-M-51—Macaroni, Spaghetti, and Vermicelli.
N-N-591—Noodles.
N-R-351—Rice, Milled.
EE-M-591—Monosodium Glutamate.
PP-B-221—Beef; Fresh (Chilled or Frozen).
RR-S-366—Sieves, Standard for Testing Purposes.
JJJ-O-533—Onions, Dehydrated.
PPP-B-636—Box, Fiberboard.
PPP-C-29—Canned Subsistence Items, Packaging and Packing of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and

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Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-35003—Peppers, Green and Red, Dehydrated.
MIL-P-35031—Peas, Dehydrofrozen.
MIL-M-35052—Milk, Nonfat, Dry.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-668—Minimum Sanitary Standards for Food Establishments.
MIL-STD-900—Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Canned Foods for the Armed Forces.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Governmental:

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

(Copies may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402. Prices may be obtained from the Superintendent of Documents.)

Regulations Governing Meat Inspection of the U. S. Department of Agriculture (USDA).

(Copies may be obtained from the Meat Inspection Division, Consumer and Marketing Service, U. S. Department of Agriculture, Washington, D. C. 20250.)

Regulations Governing the Inspection of Poultry and Poultry Products (7CFR Part 81).

(Copies may be obtained from the Poultry Division, Consumer and Marketing Service, U. S. Department of Agriculture, Washington, D. C., 20250.)

Regulations Governing the Inspection and Certification of Processed Fruits and Vegetables and Related Products.

Regulations Governing the Inspection and Certification of Fresh Fruits and Vegetables and Related Products.

United States Standards for Grades of Butter.

United States Standards for Condition of Food Containers.

United States Standards for Grades of (Canned, Frozen or Fresh as applicable):

Beans, green and wax	Okra
Beans, lima	Peas
Butter	Peppers, sweet
Corn, whole kernel	Tomato juice
Carrots	Tomato paste
Mushrooms	Tomato puree

(Single copies may be obtained, without charge, from the U. S. Department of Agriculture, Washington, D. C., 20250.)

Nongovernmental:

Official Methods of Analysis of the Association of Official Analytical Chemists.

(Copies may be obtained from the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D. C., 20044.)

Official and Tentative Methods of the American Oil Chemists Society.

(Copies may be obtained from the American Oil Chemists Society, 35 E. Wacker Drive, Chicago, Illinois, 60601.)

3. REQUIREMENTS

3.1 General all styles, types and forms. All canned soups shall be prepared from good quality, clean, sound vegetable, spices, and other materials peculiar to the variety of soup desired. All meat or meat products

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shall be prepared only in an establishment regularly operated under the supervision of the Meat Inspection Division in accordance with the Regulations Governing Meat Inspection of the USDA. All poultry or poultry products shall be prepared only in an establishment regularly operating under the supervision of the Poultry Division of the USDA in accordance with Regulations Governing the Inspection of Poultry and Poultry Products.

3.2 Workmanship. The canned soups shall be processed under sanitary conditions and in accordance with good commercial practice. The product shall be sufficiently processed by heat to assure preservation in hermetically sealed containers as specified in the invitation for bids. Containers shall be sound and clean. Cans shall be free from rust and serious dents, and the ends shall be flat or concave.

3.2.1 *Military agencies.* In addition to meeting the requirements of 3.1 and 3.2, all products other than those containing beef or poultry shall be prepared and processed with minimum delay between the various stages of production in establishments meeting the requirements of MIL-STD-668.

3.3 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and general regulations for its enforcement.

3.4 Fill of container. The containers shall be filled with soup as full as practicable without impairment of quality.

3.5 Soup styles.

3.5.1 *Style A—standard market variety.* Standard market variety of forms a and b soups after reconstitution in accordance with 5.2.1, shall have a characteristic good body, aroma, flavor, color, and consistency, typical of the soup specified.

3.5.2 *Style B—special formula.* Components shall meet the requirements of table I. Except as noted in table I, the vegetable components shall be derived from fresh, frozen, or canned sources or any combination thereof. Frozen and canned vegetables and butter shall be of a quality equivalent to U. S. grade B (extra standard), or better, of the respective U. S. Standard for Grades. When fresh vegetables are used, they shall be prepared in such a manner that they are comparable in quality to the frozen product meeting the requirements of U. S. grade B (extra standard), or better, of the U.S. Standards for Grades of the particular frozen ingredient. Inspection and certification for processed and fresh vegetable items shall be in accordance with the Regulations Governing the Inspection and Certification of Processed Fruits and Vegetables and Related Products, and the Regulations Governing the Inspection and Certification of Fresh Fruits and Vegetables and Related Products.

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TABLE I

Ingredients	~~Applicable type, style,~~ class, color, size	~~Minimum requirements, equivalent quality~~ on ingoing basis
Barley, pearl	Color I, white.	Commodity requirements of N-B-121.
Beans, green	Round types, styles other than sliced lengthwise. Pieces not greater than 1/2 inch in length.	
Beans, lima	Beans shall be of such size as to pass through a sieve containing 5/8" round holes.	
Beef	Approximately 3/8" dice.	Commodity requirements of PP-B-821 of any style, class, grade, or cutout category except ground beef, and except that the combination surface and seam fat shall not exceed 1/4 inch in thickness at any point.
Butter		United States Standards for Grades of Butter.
Carrots	Approximately 3/8" dice.	
Celery, fresh	Color—white (blanched) or ~~green varieties. Pieces not~~ greater than 1/2 inch in any dimension.	Cleaned, trimmed, sorted, and washed for removal ~~of black heart, brown stem, decay pithy branches,~~ disease, seed stems, insects, and mechanical or other damage.
Celery, dehydrated	Pieces not greater than 1/2 inch in any dimension.	Product rehydrated by adding 1.0 avoirdupois ounce of dehydrated product to 20 fluid ounces of cold water in a cooking vessel, covering, bringing quickly to a boil, and simmering for 20 minutes shall possess color, texture, flavor, and odor typical of properly prepared diced cooked celery.
Chicken meat	Approximately 3/8" dice.	Eviscerated chickens shall be cooked in water in a sanitary manner with no added ingredients, until the meat can be easily removed from the bones. Not more than 4 batches of chickens may be cooked in the same broth. Frozen chickens may be used provided they have not been held in the freezer (0°F. or below) more than 180 days, show no evidence of deterioration, and are defrosted in a strictly sanitary manner.
Chicken fat	Fat from broth.	Chicken fat obtained from the broth may be used ~~without rendering, provided it has not been held~~ for a period exceeding 18 hours or at a temperature exceeding 40°F.
	Rendered fat.	Chicken fat shall be the product obtained by rendering leaf fat of fowl and shall have been prepared in a plant operating under the supervision of the Poultry Division, Consumer and Marketing Service, USDA. Fat shall not have been stored ~~longer than 90 days nor at a temperature greater~~ than 40°F. after removal from the fowl. The rendering shall have been accomplished at a temperature of 240° to 250° F., and the total heating period shall not have exceeded 2-1/2 hours. Contact of the chicken fat with exposed copper, copper containing or iron surfaces shall be avoided at all times. An antioxidant of the types and in the amount approved by the Meat Inspection Division, Consumer and Marketing Service, USDA, and the Federal Food and Drug Administration may be incorporated into the

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TABLE I (*cont'd*)

Ingredients	Applicable type, style, class, color, size	Minimum requirements, equivalent quality on ingoing basis
Chicken fat (cont'd)	Rendered fat. (cont'd.)	rendered chicken fat. The free fatty acid content of the rendered chicken fat shall not exceed 1.2 percent, calculated as oleic acid, and the peroxide value shall not exceed 2.0 milliequivalents (meq.) of peroxide per kilogram (kg.) of fat. The fat shall be in prime condition at the time of use, shall be free from foreign or off flavors and odors, and shall have a characteristic mild chicken-fat odor.
Corn, sweet	Golden or yellow varieties, whole kernel.	
Cream and half-and-half, fresh		Commodity requirements of C-C-671.
Macaroni, alphabet	Group I, type A, class 1, style a.	Commodity requirements of the respective group, type, class, and style of N-M-51 except that the macaroni shall be in the form of letters of the alphabet normally used in canned soups and the moisture requirement may be disregarded if fresh macaroni is used.
Milk, evaporated		Commodity requirement of C-M-371.
Milk, fresh, whole		Commodity requirements of C-M-381.
Milk, nonfat, dry	Type I, style B.	Commodity requirements for the respective type and style of MIL-M-35052 or C-M-350.
Monosodium glutamate		Commodity requirements of EE-M-591.
Mushrooms, canned	Type (a), white or cream color. Pieces not greater than 3/8 inch in any dimension.	Size, style, and symmetry requirements of U. S. Grades are waived.
Mushrooms, fresh	White color. Pieces not greater than 3/8 inch in any dimension.	Cleaned, trimmed, sorted, and washed to remove insect or pathological damage.
Noodles	Type I, class A, 1/4 ± 1/16 inch wide, 0.033 ± 0.005 inch thickness, and 4 ± 2 inches long.	Commodity requirements N-N-591 except for moisture in case of fresh noodles.
Okra	Unfermented type. Pieces not greater than 1/2 inch in length.	Length of unit requirements of U. S. Standard for grades are waived.
Onions, fresh	Sweet Spanish, yellow globe, white globe varieties.	Cleaned, trimmed, peeled, sorted, washed, and minced or chopped.
Onions, dehydrated	Type I, II, or V.	Commodity requirements for the respective type of JJJ-O-522.
Peas		Dehydrofrozen peas shall meet the commodity requirements of MIL-P-35031.
Peppers, sweet	Pieces not greater than 1/4 inch in any dimension.	Dehydrated red peppers shall comply with the commodity requirements of type II, MIL-P-35003, except for piece size.
Potatoes, white, fresh	Approximately 3/8" dice.	Cleaned, peeled, trimmed, sorted, and washed. Potatoes shall be of such type and condition that they will not disintegrate, discolor, or become mushy or soggy after processing.
Rice, milled	Type I. Light cream to white.	Commodity requirements for the respective type of U. S. No. 2 grade or better of N-R-351.
Salt	Noniodized refined sodium chloride with or without anticaking agent.	

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TABLE I (*cont'd*)

Ingredients	Applicable type, style, class, color, size	Minimum requirements, equivalent quality on ingoing basis
Spices	Whole ground, or soluble or essential oils.	
Sweetening		The sweetening ingredients shall be sugar (sucrose, liquid sugar, invert sugar sirup, any combination thereof, or a mixture of one or more of the aforementioned ingredients with dextrose or corn sirup, (liquid or dry) or both, in such proportions as will result in the finished product meeting the requirements of 3.6). The sweetening ingredients shall meet the requirements of MIL-STD-900.
Thickening ingredients cornstarch		Commodity requirements of N-C-541 and MIL-STD-900.
Flour	Type I or II, class A.	Commodity requirements of N-F-481 and MIL-STD-900.
Waxy-maize or waxy-sorghum starch		Waxy-maize or waxy-sorghum starch shall be the finely powdered starch product extracted from waxy-maize or waxy-sorghum, respectively. They shall be white, contain not more than 12 percent by weight, of moisture, and comply with MIL-STD-900. The starch shall contain not more than 12 percent, by count, of blue staining cornstarch as indicated by the method given in 4.3.1.2. A 7 percent by weight, starch-water paste heated to 195° F., cooled, and allowed to stand for 24 hours at 40°F., shall be clear, cohesive, and not gelled.
Tomato products		In accordance with applicable USDA Grade Requirements.

3.5.2.1 *Special formula soups.* Special formula soups shall be of form a (ready to serve), or form b (condensed), and meet the requirements for the respective soup listed as follows:

3.5.2.1.1 *Type I—vegetable with beef.*

3.5.2.1.1.1 *Ingredients.* The ingredients shall be prepared in accordance with the following formula limitations as expressed in pounds of material per 100 pounds of formula weight. (Note: After the proper amounts of salt, tomato solids, and beef stock or broth are added, 200 pounds of the basic ingredients shown below should make approximately 60 gallons of ready to serve soup or 30 gallons of condensed soup.)

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	Maximum (pounds)			Minimum (pounds)	
Potatoes (fresh) ...	30.0		Total not to exceed 54.0	—	
Carrots (fresh) ...	26.0			—	
Barley, pearl	1.5	Total not to exceed 5.25		—	
Rice	1.5			—	
Alphabet (dry)	4.0			—	
Macaroni, spaghetti, vermicelli .					
Beef (uncooked basis)	—			20.0	
Onions (fresh) ...	3.0			1.0	
Peas (frozen) ..	7.85			3.0	
Corn (frozen) ..	7.85			—	
Celery (fresh) ...	3.0			1.0	Total not less than 26.0
Green beans (frozen) ..	6.0			—	
Lima beans (frozen) ..	5.0			—	
Okra (fresh)	2.0			—	
Sweet red peppers (fresh) ...	2.0			1.0	

The following may be substituted for the relevant ingredients shown in the above formula provided they are used in a manner and in proportionate amount without adversely affecting the finished product:

Carrots—frozen or canned.	Corn—canned or fresh.
Alphabet noodles—freshly prepared.	Celery—dehydrated.
Onions—dehydrated (chopped).	Green beans—canned or fresh.
Peas—canned or fresh.	Lima beans—canned or fresh.
Okra—frozen or canned.	Sweet red peppers—dehydrated frozen, or brined.

3.5.2.1.1.1.1 *Tomato solids, salt-free.* Salt-free tomato solids, supplied as tomato juice, puree, or paste, or a combination thereof, shall be added in the ration of not less than 8.0 lbs. to 60 gallons of ready to serve soup or 30 gallons of condensed soup.

3.5.2.1.1.1.2 *Beef stock (or broth).* Beef stock or broth shall have been prepared from the split bones of beef, meat from plates or briskets, oxtails, (or any combination thereof), and potable water, to result in a nonfat salt-free solids content of not less than 1.0 percent, by weight, for form a, and 2.0 percent, by weight, for form b.

3.5.2.1.1.1.3 *Spices, monosodium glutamate, and sweetening ingredients.* Spices, monosodium glutamate, and sweetening ingredients may be added in quantities such as to enhance but not mask the natural vegetable-beef soup flavor.

3.5.2.1.1.1.4 *Salt.* Salt shall be added in quantities necessary to meet the requirements of table I.

3.5.2.1.1.1.5 *Thickening ingredients.* Thickening ingredients may be used provided the quantity added does not adversely affect the consistency of the finished product.

3.5.2.1.1.2 *Finished soup.* The finished vegetable with beef soup shall conform to table II.

TABLE II. *Vegetable with beef soup*

Requirements	Form a (percent by weight)		Form b (percent by weight)	
	Min.	*Max.*	*Min.*	*Max.*
Physical:				
Drained weight of net contents (average of all sample cans).	30.0	—	60.0	—
Drained weight of net contents (individual cans).	22.5	—	54.0	—
Analytical:				
Salt (calculated as NaCl).	1.0	1.25	2.0	2.5
Total fat.	—	2.0	—	4.0

3.5.2.1.1.2.1 *Drained weight.* The finished soup shall contain a sufficient quantity of the basic ingredients in proportions as specified in 3.5.2.1.1.1, to assure a finished canned soup meeting the requirements of

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table II. Drained weight shall be determined as specified in 4.3.3.

3.5.2.1.1.2.2 *Color, odor, flavor, and texture.* The finished soup, when prepared for serving as specified in 5.2.1, shall possess a typical, vegetable-beef soup color, odor, and flavor; the soup shall be free of scorched, rancid, or other objectional flavors. Vegetables shall be well cooked but not mushy. The beef shall be well cooked and shall be soft and tender but not extremely mushy.

3.5.2.1.2 *Type II. Chicken.*

3.5.2.1.2.1 *Class 1. Chicken-noodle.*

3.5.2.1.2.1.1 *Ingredients.*

3.5.2.1.2.1.1.1 *Chicken broth.* Chicken broth shall have been prepared from whole eviscerated chicken and potable water in such proportions as to obtain a broth with a nonfat salt-free solids content of not less than 1.25 percent, by weight, for form a and 2.5 percent by weight, for form b. Not more than 4 batches of chickens may be cooked in any single broth.

3.5.2.1.2.1.1.2 *Chicken meat.* Chicken meat shall be added in quantities to meet the requirements of table III.

3.5.2.1.2.1.1.3 *Egg noodles.* Egg noodles shall be added in quantities to meet the requirements of table III.

3.5.2.1.2.1.1.4 Spices, celery, onions, or sweetening ingredients (or a combination thereof), may be added in quantities such as to enhance but not mask the natural chicken soup.

3.5.2.1.2.1.1.5 *Monosodium glutamate.* Monosodium glutamate shall be added in quantities of not less than 0.15 percent, by weight, for form a, and 0.25 percent, by weight for form b soups.

3.5.2.1.2.1.1.6 *Salt.* Salt shall be added in quantities necessary to meet the requirements of table III.

3.5.2.1.2.1.1.7 *Chicken fat.* Chicken fat shall be added in quantities to meet the requirements of table III.

3.5.2.1.2.1.1.8 *Waxy-maize or waxy-sorghum starch.* Waxy-maize or waxy-sorghum starch in quantities up to 0.5 percent, by weight, may be added.

3.5.2.1.2.1.1.9 *Thickening ingredients.* Thickening ingredients may be used provided the quantity added does not adversely affect the consistency of the finished product.

3.5.2.1.2.1.2 *Finished soup.* The finished chicken-noodle soup shall conform to table III.

TABLE III. *Chicken-noodle soup*

Requirements	Form a (percent by weight)		Form b (percent by weight)	
	Min.	Max.	Min.	Max.
Physical:				
Drained weight of net contents (average of all sample cans).	90.0	—	90.0	—
Drained weight of net contents (individual cans.)	22.5	—	54.0	—
Weight of chicken meat in net contents (individual cans).	2.0	—	4.0	—
Analytical:				
Total solids content.	8.0	—	15.0	—
Salt (calculated as NaCl).	1.0	1.25	2.0	2.50
Total fat.	1.25	2.0	2.5	4.0

3.5.2.1.2.1.1 *Drained weight.* Drained weight shall be determined as specified in 4.3.3 and 4.3.4.

3.5.2.1.2.1.2.2 *Color, odor, flavor and texture.* The finished soup, when prepared for serving as specified in 5.2.1, shall possess a typical yellow color and characteristic chicken-noodle soup odor and flavor. The soup shall be free of scorched, rancid, or other objectionable flavors. Noodles shall

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be well cooked and tender, but not mushy. Chicken meat shall be firm and tender.

3.5.2.1.2.2 *Class 2. Chicken-rice.* Chicken-rice soup shall conform to 3.5.2.1.2.1 except that rice shall be used in lieu of egg noodles, and the drained weight and the percent chicken meat content shall meet the requirements of table IV when determined in accordance with 4.3.3.

TABLE IV. *Chicken-rice soup*

Requirements	Form a (percent by weight)		Form b (percent by weight)	
	Min.	*Max.*	*Min.*	*Max.*
Physical:				
Drained weight of net contents (average of all sample cans).	20.0	—	40.0	—
Drained weight of net contents (individual cans).	15.0	—	36.0	—
Weight of chicken meat in net contents (individual cans) (see 4.3.4) (total solids, total fat and salt as in table III).	2.0	—	4.0	—

3.5.2.1.3 *Type III. Cream.*

3.5.2.1.3.1 *Class 1. Cream-of-tomato.*

3.5.2.1.3.1.1 *Ingredients.*

3.5.2.1.3.1.1.1 *Added fat.* Added fat shall be supplied from butter, cream, half-and-half, fresh whole milk, evaporated milk, or a combination of any of these.

3.5.2.1.3.1.1.2 *Milk, nonfat solids.* Nonfat milk solids derived from fresh milk, nonfat dry milk solids, evaporated milk, fresh cream, half-and-half or combinations thereof shall be added in a quantity of not less than 1.6 percent, by weight, for form a, and 3.2 percent, by weight, for form b soups.

3.5.2.1.3.1.1.3 *Tomato solids, salt-free.* Salt-free tomato solids shall be supplied as tomato juice, puree, or paste, or a combination of any of these in a quantity of 8.2 ± 0.2 percent, by weight, for form a, and 6.4 ± 0.4 percent, by weight, for form b soups.

3.5.2.1.3.1.1.4 Spices, monosodium glutamate, onion powder, and sweetening ingredients (or a combination thereof), may be added in quantities such as to enhance but not mask the characteristic tomato soup flavor.

3.5.2.1.3.1.1.5 *Thickening ingredients.* Thickening ingredients in the form of cornstarch or wheat flour (or a combination of both), shall be used in proportions to produce a finished soup with an apparent viscosity as specified in table V and determined as specified in 4.3.5.

3.5.2.1.3.1.1.6 *Salt.* Salt shall be added in quantities necessary to meet the requirements of table V.

3.5.2.1.3.1.2 *Finished soup.* The finished cream-of-tomato soup shall conform to table V.

TABLE V. *Cream-of-tomato soup*

Requirements	Form a (percent by weight)		Form b (percent by weight)	
	Min.	*Max.*	*Min.*	*Max.*
Physical:				
Apparent viscosity.	22.0 ± sec. per 100 revolutions, using an 80-g. weight (std. as specified in 4.3.5.2 and 4.3.5.2.1)		17.0 ± 4.0 sec. per 100 revolutions, using a 400-g. weight (std. as specified in 4.3.5.2 and 4.3.5.2.1)	
Analytical:				
Total fat.	2.0	3.0	3.0	4.0
Salt (calculated as NaCl).	1.0	1.25	2.0	2.5

3.5.2.1.3.1.2.1 *Consistency.* The finished soup shall be of a smooth consistency with no indication of curdling or lumping.

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3.5.2.1.3.1.2.2 *Color, odor, and flavor.* The finished soup, when prepared for serving as specified in 5.2.1, shall possess a bright, typical cream-of-tomato soup color, odor, and flavor, and shall be free of scorched, rancid, or other objectionable flavors.

3.5.2.1.3.2 *Class 2. Cream-of-mushroom.*

3.5.2.1.3.2.1 *Ingredients.*

3.5.2.1.3.2.1.1 *Mushrooms.* Mushrooms shall be added in quantities to meet the requirements of table VI.

3.5.2.1.3.2.1.2 *Added fat.* Added fat shall be supplied from butter, cream, half-and-half, fresh whole milk, evaporated milk, or a combination of any of these.

3.5.2.1.3.2.1.3 *Milk, nonfat, solids.* Nonfat milk solids derived from fresh milk, nonfat dry milk solids, evaporated milk, fresh cream, half-and-half, or combinations thereof, shall be added in a quantity of not less than 2.1 percent, by weight, for form a, and 4.0 percent, by weight, for form b soup.

3.5.2.1.3.2.1.4 Spices or onion powder or a combination thereof, may be added in quantities such as to enhance but not mask the characteristic mushroom soup flavor.

3.5.2.1.3.2.1.5 *Thickening ingredients.* Thickening ingredients in the form of cornstarch or wheat (or a combination of both), shall be used in proportions to produce a finished soup with an apparent viscosity as specified in table VI and determined as specified in 4.3.5.

3.5.2.1.3.2.1.6 *Monosodium glutamate.* Monosodium glutamate shall be added in quantities of not less than 0.15 percent, by weight, for form a, and 0.25 percent, by weight, for form b soups.

3.5.2.1.3.2.1.7 *Salt.* Salt shall be added in quantities necessary to meet the requirements of table VI.

3.5.2.1.3.2.2 *Finished soup.* The finished cream-of-mushroom soup shall conform to table VI.

TABLE VI. *Cream-of-mushroom soup*

Requirements	Form a (percent by weight)		Form b (percent by weight)	
	Min.	*Max.*	*Min.*	*Max.*
Physical:				
Drained weight of mushrooms of net contents (average of all sample cans).	8.5	—	17.0	—
Drained weight of mushrooms of net contents (individual sample cans).	6.5	—	15.3	—
Apparent viscosity.	19.0 ± 4.0 sec. per 100 revolutions, using a 100-g. weight (std. as specified in 4.3.5.2 and 4.3.5.2.1)		24.0 ± 4.0 sec. per 100 revolutions, using a 150-g. weight (std. as specified in 4.3.5.2 and 4.3.5.2.1)	
Analytical:				
Total fat.	2.0	—	6.0	—
Salt (calculated as NaCl).	1.0	1.25	2.0	2.5

3.5.2.1.3.2.2.1 *Drained weight.* The average weight of mushrooms determined as specified in 4.3.3 shall meet the requirements of table VI.

3.5.2.1.3.2.2.2 *Consistency.* The finished soup shall be of a smooth consistency except for mushroom pieces, with an absence of curdling and lumping.

3.5.2.1.3.2.2.3 *Color, odor, and flavor.* The finished soup, when prepared for serving as specified in 5.2.1, shall possess a cream to light tan color; shall possess a typical cream-of-mushroom soup odor and flavor, and shall be free of scorched, rancid, or other objectionable odors and flavors.

3.5.2.1.3.3 *Class 3. Cream-of-pea soup.*

3.5.2.1.3.3.1 *Ingredients.*

3.5.2.1.3.3.1.1 *Pea solids.* Pea solids shall be supplied as canned, fresh, frozen, or dehydrofrozen peas at a rate of not less than 6.4 percent, by weight, for form a, and 12.8 percent, by weight, for form b soups.

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3.5.2.1.3.3.1.2 *Added fat.* Added fat shall be supplied from butter, cream, half-and-half, fresh whole milk, evaporated milk, or a combination of any of these.

3.5.2.1.3.3.1.3 *Milk, nonfat, solids.* Nonfat milk solids derived from fresh milk, nonfat dry milk solids, evaporated milk, fresh cream, half-and-half, or combinations thereof, shall be added in a quantity of 1.8 ± 0.3 percent by weight, for form a, and 3.6 ± 0.6 percent, by weight, for form b soups.

3.5.2.1.3.3.1.4 Spices, monosodium glutamate, onion powder, ground celery, or sweetening ingredients (or a combination thereof), may be added in such quantities as to enhance but not to mask the characteristic pea soup flavor.

3.5.2.1.3.3.1.5 *Thickening ingredients.* Thickening ingredients in the form of cornstarch or wheat flour (or a combination of both), may be added to obtain the required apparent viscosity, provided that all other ingredient requirements are met.

3.5.2.1.3.3.1.6 *Salt.* Salt shall be added in quantities necessary to meet the requirements of table VII.

3.5.2.1.3.3.2 *Finished soup.* The finished cream-of-pea soup shall conform to table VII.

TABLE VII. *Cream-of-pea soup*

Requirements	Form a (percent by weight)		Form b (percent by weight)	
	Min.	Max.	Min.	Max.
Physical:				
Apparent viscosity.	27.0 ± 3.0 sec. per 100 revolutions, using a 70-g. weight (std. as specified in 4.3.5.2 and 4.3.5.2.1)		Same as a, after being reconstituted.	
Analytical:				
Salt (calculated as NaCl).	1.0	1.25	2.0	2.50
Total fat.	3.0	—	6.0	—

3.5.2.1.3.3.2.1 *Consistency.* The finished soup shall be of a smooth consistency with an absence of curdling and lumping.

3.5.2.1.3.3.2.2 *Color, odor, and flavor.* The finished soup, when prepared for serving as specified in 5.2.1, shall possess an odor, flavor, and green color typical of cream-of-pea soup, and shall be free of scorched, rancid, or other objectionable odors and flavors.

3.5.2.1.3.4 *Class 4. Cream-of-chicken.*

3.5.2.1.3.4.1 *Ingredients.*

3.5.2.1.3.4.1.1 *Chicken broth.* Chicken broth shall have been prepared from whole eviscerated chickens and potable water, in such proportions as to obtain a broth with a nonfat salt-free solids content of not less than 1.25 percent, by weight, for form a, and 2.5 percent, by weight, for form b. Not more than 4 batches of chicken shall be cooked in any single broth.

3.5.2.1.3.4.1.2 *Chicken meat.* Chicken meat shall be added in quantities to meet the requirements of table VIII.

3.5.2.1.3.4.1.3 *Milk, nonfat, solids.* Nonfat milk solids derived from fresh milk, nonfat dry milk solids, evaporated milk, fresh cream, half-and-half or combinations thereof, shall be added in a quantity of not less than 1.0 percent, by weight, for form a, and 2.0 percent, by weight, for form b soups.

3.5.2.1.3.4.1.4 *Thickening ingredients.* Thickening ingredients in the form of cornstarch or wheat flour, or a combination of both, shall be used in proportions to produce a finished soup with an apparent viscosity as specified in table VIII and determined as specified in 4.3.5.

3.5.2.1.3.4.1.5 *Monosodium glutamate.* Monosodium glutamate shall be added in quantities of not less than 0.15 percent, by weight, for form a, and 0.25 percent, by weight, for form b soups.

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3.5.2.1.3.4.1.6 *Salt.* Salt shall be added in quantities necessary to meet the requirements of table VIII.

3.5.2.1.3.4.1.7 *Chicken fat.* Chicken fat shall be added in quantities to meet the requirements of table VIII.

3.5.2.1.3.4.1.8 Spices, powdered or minced onion, carrots, celery, and sweetening ingredients (or a combination thereof), may be added in quantities such as to enhance but not mask the characteristic cream-of-chicken soup flavor.

3.5.2.1.3.4.2 *Finished soup.* The finished cream-of-chicken soup shall conform to table VIII.

TABLE VIII. *Cream-of-chicken soup*

Requirements	Form a (percent by weight)		Form b (percent by weight)	
	Min.	Max.	Min.	Max.
Physical:				
Drained weight of net contents	2.5	—	5.0	—
Weight of chicken meat of net contents.	2.0	—	4.0	—
Apparent viscosity.	20.5 ± 2.5 sec. per 100 revolutions, using a 100-g. weight (std. as specified in 4.3.5.2 and 4.3.5.2.1)		18.5 ± 2.0 sec. per 100 revolutions using a 100-g. weight (std. as specified in 4.3.5.2 and 4.3.5.2.1)	
Analytical:				
Salt (calculated as NaCl).	1.0	1.25	2.00	2.50
Total fat.	2.50	3.50	5.00	7.00

3.5.2.1.3.4.2.1 *Drained weight.* Drained weight of net contents and weight of chicken meat shall be determined in accordance with 4.3.5 and 4.3.4.

3.5.2.1.3.4.2.2 *Color, odor, flavor, and consistency.* The finished soup, when prepared for serving as specified in 5.2.1, shall be of a smooth consistency (except for chicken, celery, onion, and carrot pieces), with an absence of curdling and lumping. The product shall possess a typical cream-of-chicken soup color, odor, and flavor, and shall be free of scorched, rancid, or other objectionable odors and flavors.

3.6 Workmanship.

3.6.1 *Civil agencies.* The plant, grounds, buildings, equipment, personnel practices, packaging materials, sanitary conditions, and practices used in the production of the product, shall be such as to minimize the possibility of contamination of the product through microbial growth, dust, pests, rodent excreta, or other foreign matter. Each product shall be processed with a minimum of delay between various steps of production. The product shall be packaged under modern sanitary conditions.

3.6.2 *Military agencies.* The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared and processed with minimum delay between the various stages in production.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. Sampling of finished product covered by this specification, and samples of any materials, components, or constituents entering into the preparation of the finished product may be taken at any time by the Government inspector for inspection purposes.

4.2 Inspection.

4.2.1 *General.* Inspection shall be made by the procuring agency or duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing and packaging, or at any suitable point in transit or after delivery to the point of destination.

4.2.2 *Factory inspection.* The passing as satisfactory of any detail of processing of materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final inspection.

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4.2.3 *Final inspection.* Unless otherwise specified (see 6.1), final inspection shall be made after delivery to the point of destination.

4.2.4 *Quality assurance provisions.*

4.2.4.1 *Responsibility for inspection.* The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified (see 6.1) in the contract or order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.2.4.1.1 Inspection records of the examinations and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.4.2 *Preacceptance inspection (plant sanitation inspection).*

4.2.4.2.1 *Civil agencies.* Inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.4.2.2 *Military agencies.* Award of contracts for styles A and B shall be limited to:

	Plants
Type I	Operating under the continuous inspection supervision of the Meat Inspection Division, Consumer and Marketing Service of the USDA.
Type II: Classes 1 and 2	Operating under the continuous inspection of the Poultry Division, Consumer and Marketing Service of the USDA.
Type III: Class 4	
Type III: Classes 1, 2 and 3	That have received prior sanitary approval by the Military Veterinary Services in accordance with MIL-STD-668.

4.2.4.3 *Sampling for inspection.* Sampling for inspection for style B soups shall be in accordance with provisions of MIL-STD-105, except as indicated herein.

4.2.4.3.1 *Component and material inspection.* In accordance with 4.2.4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification.

4.2.4.3.1.1 *Examination of ingredients.*

4.2.4.3.1.1.1 *Examination of beef; pearl barley; celery dehydrated; nonfat dry milk; corn starch; flour; monosodium glutamate; noodles; macaroni; dehydrated onions; dehydrated red peppers; dehydrofrozen peas; and evaporated milk.* Examination of the above ingredients shall be in accordance with the examination criteria of product characteristics of pertinent specifications and additional requirements, referenced in table I, when applicable. The composite sample(s) for each ingredient, except beef, shall be derived from five primary contain-

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ers in a lot. If a lot consists of less than 5 primary containers, the composite sample shall be extracted and examined from each of the primary containers. Use of ingredient(s) having one or more defects shall be cause for rejection of the involved quantity of the finished product.

Ingredient	Sample
Pearl barley	8 ounce composite.
Nonfat dry milk	8 ounce composite (2 oz. composite for rehydration)[1].
Cornstarch	4 ounces.
Flour	8 ounces.
Monosodium glutamate	2 ounces.
Noodles	8 ounce composite.
Dehydrated onions	4 ounce composite[2] (4 oz. composite for rehydration)[1]—1 ounce[3].
Macaroni	8 ounces.
Dehydrefrozen peas	8 ounces.
Dehydrated red pepper	4 ounce composite (1 oz. composite for rehydration)[1].
Evaporated milk	8 fluid ounces.
Celery dehydrated	4 ounce composite.

[1] The composite sample shall be rehydrated as specified in the pertinent subsidiary specification, and shall be derived from the sample used for dehydration examination.
[2] Applicable to chopped minced onions.
[3] Applicable to powdered product.

4.2.4.3.1.2 *Examination of cream, and half-and-half, fresh; and milk, whole fresh.* Examination of cream and half-and-half fresh; and milk, whole, fresh, as concerns approved source of supply, shall be ascertained by examination of labels, certificates, approved tests, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary, to determine condition of the product, e.g., off flavors and odors, presence of foreign materials such as metal, wood, glass, insect, and insect fragments, filth, etc. The sample unit for examination shall be 1 pint of product from a primary container. Five sample units shall be examined. If a lot consists of less than 5 primary containers, a sample unit shall be extracted and examined from each primary container. Use of an ingredient not conforming to the source requirements or one or more sample units not conforming to condition requirements, shall indicate an unacceptable ingredient, and use of such shall be cause for rejection of the involved finished product made therefrom.

4.2.4.3.1.3 *Examination of fresh green beans, fresh and dehydrated celery, fresh peas, fresh lima beans, fresh carrots, fresh corn, fresh okra, fresh or brined red sweet peppers, and fresh tomatoes.* Examination of the above fresh ingredients (after processing), shall be in accordance with the United States Standards for Grade of the comparable frozen or canned product, except processed color or flavor to the extent applicable except tomatoes fresh, which shall be examined in accordance with standards for grades of the USDA. A three pound sample of each ingredient shall be extracted and graded. Use of any ingredients not comparable to the required grade requirements of the above referenced standards shall be cause for rejection of the involved finished product made therefrom.

4.2.4.3.1.4 *Examination of green beans frozen or canned, carrots canned or frozen, lima beans frozen or canned, sweet corn frozen or canned, okra frozen or canned, peas frozen or canned, sweet red peppers frozen or canned, mushrooms fresh or canned, tomato juice, tomato puree and tomato (all) canned, butter, fresh onions, chicken meat, fresh potatoes, salt, sweetening ingredients, antioxidants, rice, waxy-maize or waxy-sorghum, and spices.* Conformance of ingredients to requirements for identity (grade, size, type, formulation varieties, as applicable), storage limitations, source inspection of chicken and temperature requirements as referenced in table I, shall be ascertained by examination of labels, invoices, grade certificates, storage and temperature records, or other valid documents. In addition, all frozen ingredients shall be examined organoleptically, as nec-

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essary, prior to use to determine spoilage. Use of ingredients not conforming to one or more of the above referenced requirements shall be cause for rejection of the involved finished product.

4.2.4.3.2 *Testing of components.* Testing of components described in this specification, and changes in the testing of components listed in subsidiary specifications, shall be as follows. Lot size shall be expressed in pounds, and the sample shall be a composite derived from 5 primary containers or all the containers of a lot consisting of less than 5 primary containers. Nonconformance to one or more test requirements shall be cause for rejection of the involved finished product made therefrom.

Component	Sample	Characteristics	Results reported	Test par. & requt. par.
Rendered chicken fat	1/2 pound composite	Free fatty acid (calculated as oleic acid)	Nearest 0.1 percent	Table I, 4.2.1.1
		Peroxide value	Nearest 0.1 milliequivalent	Table I, 4.2.1.1
Waxy-maize or waxy-sorghum starch	4 ounces	Moisture	Nearest 0.1 percent	Table I, 4.2.1
		Blue staining corn-starch	Nearest 1.0 percent	Table I, 4.2.1.2
Cream and-half-and-half, fresh	1 pound composite	1	1	Table I
Milk, evaporated	2 pound composite	1	1	Table I
Milk, whole, fresh	2 pound composite	1	1	Table I
Milk, nonfat dry	1 pound composite	1	1	Table I
Macaroni, spaghetti, vermicelli	1 pound composite	1	1	Table I
Needles	1 pound composite	1	1	Table I
Peas, dehydrofrozen	1 pound composite	1	1	Table I
Monosodium glutamate	2 oz. composite	1	1	Table I
Onions, dehydrated	4 ounces composite	1	1	Table I
Peppers, sweet red, dehydrated	4 ounces composite	1	1	Table I.
Cornstarch	1/2 pound composite	1	1	Table I
Flour	1 pound composite	1	1	Table I
Waxy-maize or waxy-sorghum	4 ounces composite	1	1	Table I
Sweetening ingredient	1/2 pound composite	1	1	Table I

[1] In accordance with specification or standard as applicable.

4.2.4.4 *Examination of processing.* Examination shall be performed during all stages of processing to ascertain compliance to requirements as concerns procedures (washing, cutting, trimming, sorting, cubing, etc.), controls, and formulation. Records of time, temperature, storage, and formulation shall be maintained. Nonconformance to one or more of the applicable requirements reflected by actual examination or by records of formulation, storage, or temperature

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Table	Inspection level	Sample unit	Lot size expressed in	AQL's		
				Major	Minor	Total
IX	S-3	1 primary container	primary containers	1.5	2.5	—
X	S-2	1 primary container	primary containers	1.5	—	4.0

shall be cause for rejection of the involved quantity of finished product.

4.2.4.5 *Examination of end item.* Classification of defects found during examination shall be in accordance with tables IX and X and 4.2.5 through 4.2.7. The inspection levels, sample units, expression of lot sizes, and acceptable quality levels (AQLs) for the above referenced tables shall be as shown above. The AQL for table IX shall be expressed as percent defectives and for table X as defects per hundred units (DHU).

TABLE IX. *Examination of drained weight and net weight* [1] [4]

Category		Defects
Major	Minor	
151		Unit drained weight of net contents less than specified[2] (applicable to types I and II, classes 1 and 2, and type III, classes 2 and 4).
152		Unit drained weight of chicken meat less than specified[2] (applicable to type II, classes 1 and 2, type III, class 4).
	201	More than 5 percent under net weight specified in PPP-C-29.

[1] Lot shall be rejected if sample data indicates lot average net weight or drained weight (as specified for types I and II, classes 1 and 2, type III, classes 2 and 4), is less than specified net weight or drained weight.
[2] See 4.2.3 for procedure.
[3] See 4.2.4 for procedure.
[4] Calculate to 1/8 ounce.

TABLE X. *Examination of product characteristics* [2] [3]

Category		Defects
Major[1]	Minor	*Applicable to all types and classes*
151		Not typical flavor or odor, e.g. scorched, rancid, or other objectionable flavors.
		Applicable to type I:
152		Vegetables or beef not well cooked.
	201	Vegetables or beef mushy.
	202	Beef not soft or tender.
	203	Not typical vegetable—beef soup color.
		Applicable to type II, class 1:
	204	Noodles not soft or tender or noodles mushy.
		Applicable to type II, classes 1 and 2:
153		Not a typical bright yellow color.
154		Chicken meat not firm and tender.
		Applicable to type III, classes 1, 2, 3, and 4[3] [4]*:*
155		Not a smooth consistency, e.g., curdled or lumpy.
		Applicable to type III, class 1:
156		Not bright typical cream of tomato soup color.
		Applicable to type III, class 2:
157		Not a cream or light tan color.
		Applicable to type III, class 3:
158		Not a typical green color.
		Applicable to type III, class 4:
159		Not a typical cream-of-chicken soup color.

[1] Presence of foreign material, e.g., glass, wood, metal, filth, insect, or insect fragment shall be cause for rejection of the entire lot.
[2] Examination of form b soups shall be made after the soup has been prepared for serving in accordance with instructions.
[3] For type III, class 2, presence of mushroom pieces and for type III, class 4, presence of chicken, celery, onion and carrot pieces shall be disregarded.

4.2.5 Examination of packaging, vacuum, interior, and exterior can coating and labeling shall be in accordance with the applicable examination criteria of PPP-C-29 or referenced documents cited therein.

4.2.6 *Examination of shipping container.* When shipping containers are required to be in accordance with level A or B of PPP-C-29, the examination shall be in accordance with examination criteria in appendix of PPP-B-636. In addition, the following de-

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fects shall be included in the table of examination. Major: Marking; missing, incorrect, or illegible. Minor: Arrangement and number of contents not as specified. For level C pack, examination shall be in accordance with the applicable examination criteria of PPP-C-29.

4.2.7 Sampling procedure and acceptance criteria for testing of finished product. The finished product shall be tested for salt (calculated as NaCl), total solids, total fat, and apparent viscosity as applicable to the pertinent type or class of product. Procedures of testing shall be in accordance with 4.3.1 and 4.3.5 as applicable. The sample for testing shall be a 1-quart composite for form a, and 1-pint for form b of each type or class to be tested. The composite sample shall be derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of primary containers. Results shall be reported to the nearest unit or decimal place as specified for the specific requirement. Lot shall be rejected if test results indicate nonconformance to one or more test requirements.

4.2.8 *Condition inspection of containers.* When inspection of product for quality and condition is performed by the USDA, the U. S. Standards for Condition of Food Containers shall apply. (For applicable document governing condition inspection of shipping containers for military agencies see 4.2.6.)

4.2.9 *Standby test samples.* The Government reserves the right to withdraw and withhold standby test samples of components of finished product or both (quantity of which shall not be more than twice that required by this specification), for inspection purposes. These samples, if not used, shall be returned to the vendor.

4.3 Tests.

4.3.1 *Chemical analyses.* Unless otherwise specified, chemical analyses, when required, shall be made in accordance with the following tests of the Official Methods of Analysis of the Association of Official Analytical Chemists in effect on the date of invitation for bids (see 6.1):

Test	Chapter	Section
Total solids:	Vegetable products processed.	Total solids.
Type I—tomato solids, salt-free		
Type II, classes 1 and 2		
Salt (all types)	Vegetable products processed.	Sodium chloride.
Total fat (all types)	Dairy products.	Fat in dried milk alkaline digestion is not necessary, unless milk solids are present.
Nonfat salt-free solids (chicken and beef broth)	Meat and meat products, meat extracts, and similar products.	Moisture.
Moisture (waxy-maize and waxy-sorghum starch)	Sugar and sugar products.	Moisture vacuum-drying.

4.3.1.1 *Free fatty acid and peroxide value.* Free fatty acid and peroxide value of chicken fat shall be tested in accordance with methods Ca 5a—40 and Cd 8—53, respectively, of the Official and Tentative Methods of the American Oil Chemists Society.

4.3.1.2 *Blue-staining starch.* An aqueous suspension of the cornstarch shall be stained lightly and uniformly with iodine. The waxy-maize and waxy-sorghum granules will be stained red, while the normal starch granules are stained blue. Check approximately 500 granules microscopically, to determine the percentage of blue-stained granules.

4.3.2 *Microbiological examination.* The microbiological examination of cornstarch, flour, waxy-maize or waxy-sorghum starch, or sweetening ingredients, shall be in accordance with methods specified in MIL-STD-900.

4.3.3 *Drained weights.*

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4.3.3.1 Drained weights of types I and II, classes 1 and 2 and type III, classes 2 and 4, soups shall be determined by emptying the contents of the can in such manner as to distribute the product evenly on a U. S. Standard No. 20 woven-wire cloth sieve (0.0331-inch openings), 8 inches in diameter in accordance with RR-S-366. The sieve and product remaining on the sieve shall then be immersed in water at 58° ± 2°F. to a depth just sufficient to cover the product, and withdrawn. This operation shall be repeated twice, making a total of 3 immersions. The product shall be allowed to drain for 2 minutes, with the sieve in an inclined position to facilitate drainage. The drained weight shall be the weight of the sieve and product remaining thereon, less the weight of the wet sieve. Care shall be exercised in order to assure that the matter, other than the solid portion, is not retained on the sieve. A longer wash period accompanied by slight agitation may be necessary. However, the wash procedure should be held to a minimum of time.

4.3.4 *Weight of chicken meat.* Weight of chicken meat shall be determined by separating it from other drained solid material and weighing it to the nearest 0.1 g. The chicken meat should be collected in a glass covered petri dish to reduce evaporation losses to a minimum.

4.3.5 *Apparent viscosity.* At least 48 hours shall have elapsed after processing and cooling the soup before this test is performed.

4.3.5.1 *Equipment.*

Stormer viscosimeter
Double bladed roter—Overall length 4-1/16 inches.
Length of roter blades to shank or roter 2-1/2 inches.
Overall width (outside) of one blade to (outside) of second—1-1/2 inches.
Width of each blade—1/4 inch.
Thickness of each blade—1/16 inch.
Distance of depth mark from end of blades—1.55 inches.

4.3.5.2 *Standard weights.*

Weight (grams)	Seconds per 100 revolutions in glycerin chemically pure (c.p.) at 72° to 73°F.
400	12.0
150	25.5
100	35.0
80	42.5
70	48.0

4.3.5.2.1 *Weight standardization procedure.* Standardized weights are obtained as follows: Glycerin c.p. at 72° to 73°F. shall be used as the standardizing liquid. The time required for 100 revolutions of the spatula-like blades, using the approximate weights shown above, is to be exact to 1/10 of a second. The weights shown above and those weights specified in this specification are approximate; however, the standardized equivalent weight is to be used for the apparent viscosity determination.

Example: 400 g. is shown to obtain a 12.0 second reading for glycerin c.p. If 400 g. on the instrument used gives a time reading of 11.5 seconds for 100 revolutions, a smaller weight will result in a 12.0 second reading. This smaller weight is the standardized weight to be used whenever a 400 g. weight is specified for the apparent viscosity determination.

4.3.5.3 *Form a soup.* The apparent viscosity for ready to serve soups shall be determined in the following manner:

(*a*) Mechanically stir the soup in the can for five minutes at such a rate as to avoid whipping air into the soup and to obtain thorough mixing.

(*b*) The can and its contents shall then be placed in the stand of the Stormer viscosimeter and raised to such a position that the spatula-like blades of the rotor are immersed to the indicated depth marks. The temperature of the soup shall be 72° to 73° F. when the readings are taken.

(*c*) The standardized weight (see 4.3.5.2), specified for each soup, shall be used to apply the torque to the rotor.

(*d*) Ten repeated readings shall be taken of the time required for 100 revolutions of the spatula-like blades. These read-

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ings shall not be initiated until at least 10 revolutions have occurred and shall end 100 revolutions thereafter. The timing shall be to the tenth of a second. All readings shall be completed within a 10-minute period after the stirring indicated in (*a*). The average of these 10 readings shall be considered to be the apparent viscosity. Large decreases (0.5 second), of consecutive readings are indicative of improper or insufficient stirring.

4.3.5.3.1 The type III, class 4, form a soup, in addition to instructions for other form a soups, shall be strained through a U. S. Standard No. 20 woven wire cloth sieve (0.0331-inch openings), after stirring and apparent viscosity determined on the liquid portion thereof.

4.3.5.4 *Form b soup.* The apparent viscosity for condensed soups shall be determined as follows:

4.3.5.4.1 *Type III, class* 1. Processed as in 4.3.5.3.

4.3.5.4.2 *Type III, class* 2. Empty contents of can into Waring or similar type blender, and blend at full speed for 2 minutes. Return soup to can and proceed as in 4.3.5.3 (*b*), (*c*), and (*d*) except that the 10-minute time limit required in (*d*) is not mandatory. If consecutive readings show large decreases (0.5 second) in time, continue reading until 5 consecutive readings agree within 0.1 second. The average of these 5 readings shall be considered to be the apparent viscosity.

4.3.5.4.3 *Type III, class* 3. Empty contents of can into a 1-quart container; add an equal volume of water, and stir until the solids are uniformly distributed; pour half of this diluted soup into the original can, and proceed as specified in 4.3.5.3.

4.3.5.4.4 *Type III, class* 4. The apparent viscosity shall be determined in accordance with 4.3.5.3 on the liquid portion obtained by mixing the soup with an equal volume of water, stirring (with a mechanical stirrer) for a period of 5 minutes at such rate as to avoid whipping air into the soup, and straining the soup through a U. S. Standard No. 20 sieve (0.0331-inch openings).

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.** The product shall be packaged and packed, at the level specified (see 6.1), in accordance with PPP-C-29.

5.2 **Labeling and marking.**

5.2.1 *Labeling.* Labeling shall be in accordance with PPP-C-29 and the following, "Directions for Serving", shall be included:

DIRECTIONS FOR SERVING

For ready to serve soups

EMPTY CONTENTS INTO SUITABLE CONTAINER, HEAT TO BOILING AND SERVE.

TO HEAT*

Warm open can over flame, or place can in boiling water.

For condensed soups.

EMPTY CONTENTS INTO SUITABLE CONTAINER

Stir in 1 can of water.

Heat to boiling, stirring occasionally and serve.

* If procured for operational rations, cans shall have the above additional information included.

5.2.2 *Marking.* Shipping container shall be marked in accordance with PPP-C-29.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Style, type, class, and form of product required (see 1.2).

(*c*) When other than supplier is required to perform the examination and tests (see 4.2.4.1).

(*d*) When final inspection is to be made

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other than after delivery to the point of destination (see 4.2.3).

(*e*) When chemical analyses are required (see 4.3.1).

(*f*) Applicable level of packaging and packing required (see 5.1).

6.2 Supplier's inspection. The supplier should be required to assure that the product conforms to the specification prior to the submission to the Government for final inspection.

6.3 Destination inspection. When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Soups, not frozen.
Carload minimum weight 36,000 pounds.

Motor:

Soups, not frozen.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.5 For classification of can defects, either of the following documents is suggested as a guide:

(*a*) Visual Inspection Aids A1—C1, as specified in AMS Handbook entitled "Exterior Condition of Filled Food Containers", August 8, 1953. Single copies may be obtained from the Deputy Administrator, Consumer and Marketing Services, U. S. Department of Agriculture, Washington, D.C., 20250.

(*b*) Visual Inspection Gage Set No. 33A—2D, as specified in "Standard Classification for Can Defects", December 1, 1960.

MILITARY CUSTODIANS:

Army—GL
Navy—SA
Air Force—67

Review activity:

Army—MD
Navy—MC, SA
Air Force—67

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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